

NewsBot Intelligence System

Technical Documentation | ITAI2373 Final Project

1. System Architecture

The NewsBot is a production-ready NLP pipeline designed to ingest, analyze, and serve news data. The architecture consists of a **Data Ingestion Layer** (JSON/Web APIs), a **Processing Layer** (NLTK, Spacy, Transformers), and an **Interface Layer** (Streamlit/Flask).

2. NLP Modules & Advanced Analysis

2.1 Topic Modeling (LDA)

We utilized Latent Dirichlet Allocation (LDA) via `scikit-learn`'s `LatentDirichletAllocation` to extract latent themes from the *News Category Dataset v3*. This allows for automated grouping beyond pre-defined labels.

2.2 Multilingual Pipeline

Integrated `googletrans` (version 3.1.0a0) to provide real-time translation. Headlines in Spanish, French, or German are translated to English before being passed to the sentiment engine.

3. Code Snippet: Sentiment Pipeline

```
from transformers import pipeline

# Advanced Transformer-based Sentiment
summarizer = pipeline("summarization", model="sshleifer/distilbart-cnn-12-6")

def process_article(text):
    summary = summarizer(text[:1000], max_length=100, min_length=30)[0]
    return summary['summary_text']
```

4. Conversational Interface

The system uses **SBERT (Sentence-BERT)** to generate document embeddings. When a user enters a query, the system calculates the cosine similarity between the query vector and the news database to provide the most relevant answers.